

Chapter 10 CRT - Sophia Chen

1. A GUI is visual/ graphical interface that allows the user to interact and communicate with the program. GUI's can contain components such as buttons, text fields, frames, and labels. GUI's are also event-driven.
2. Code in the program is run when an event is sent by an interaction from the user, such as a click of a button or a text input. In response to this event, the GUI executes a method called an event handler and executes the corresponding code that relates to the button clicked.
3. GUI components can not be directly added to the JFrame. A JPanel can be first added to the JFrame, and then GUI components such as buttons or labels can be added into the JPanel. The content pane is specially designed to hold GUI components.
4. A label can not respond to events, as it is static text or an image that does not usually generate action events, such as a button.
5. A GUI needs to be run from an event-driven application to make sure that each event handler has finished executing before another one begins.
6. A label displays information, like text or an image, but cannot normally be clicked or respond to an interaction from the user. A button can also display information, but unlike a label, it is able to respond to a click to trigger an action.
8. The layout of a content pane in the GUI can be controlled by adding and using borders, using a layout manager, and setting alignments according to your needs. Layout managers can help change the order of components on a content pane. Some include FlowLayout, BoxLayout, and GridLayout.
13. Numeric data typed in a text field must first be converted into a numeric variable type, such as a double or an int. This is because even when numeric data is entered, it is stored as a string. You can use a method to return the double or int value in the string text before using it in a calculation.
14. `num1 = 8.0`
15. A combo box is a better choice because it lets the user select a name from a predetermined list of choices, eliminating spelling or typing errors from the user. It is also more convenient as the user only has to click on one choice as opposed to typing the name out.